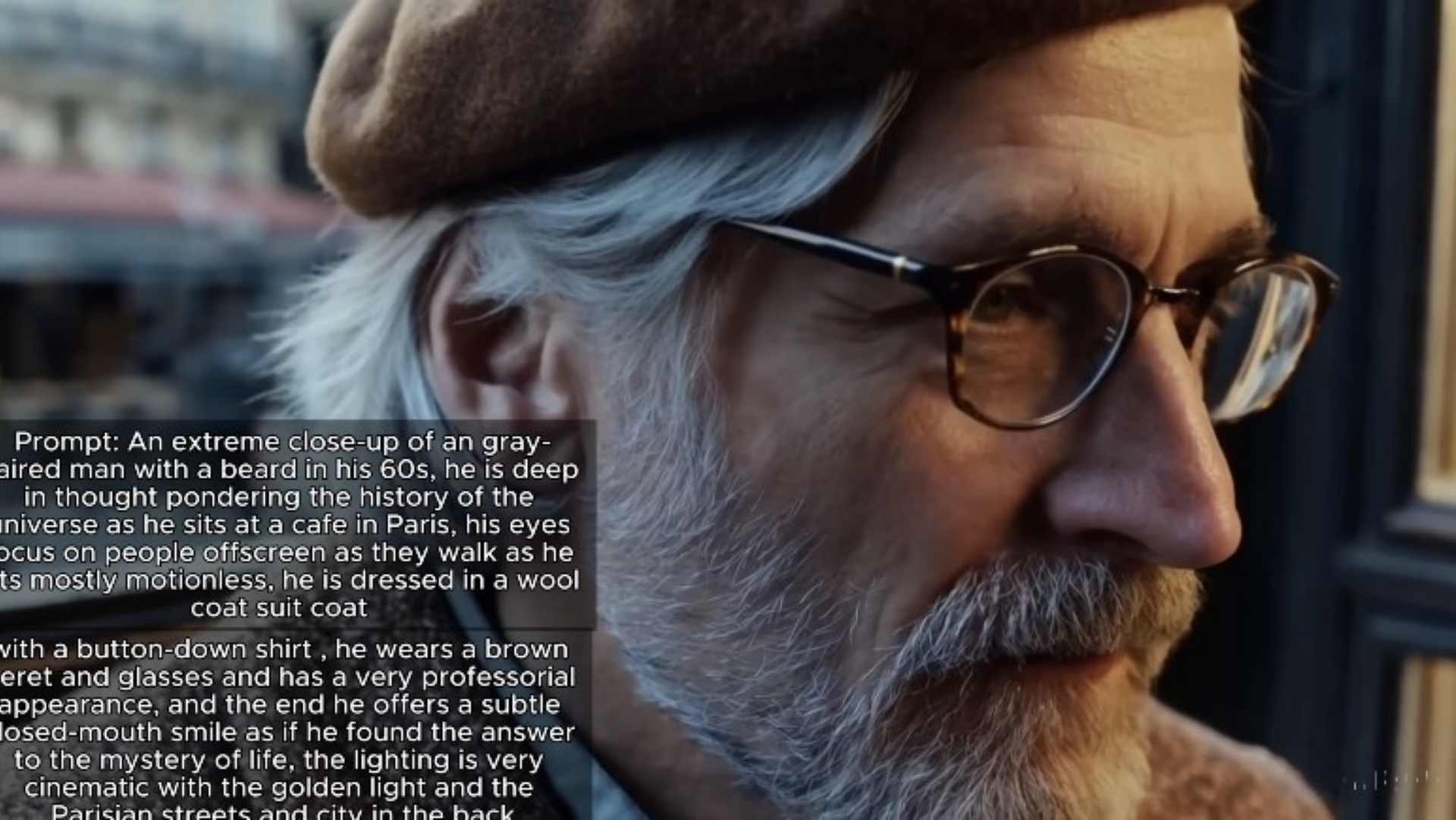
Machine Learning Foundations

Andrzej Wodecki

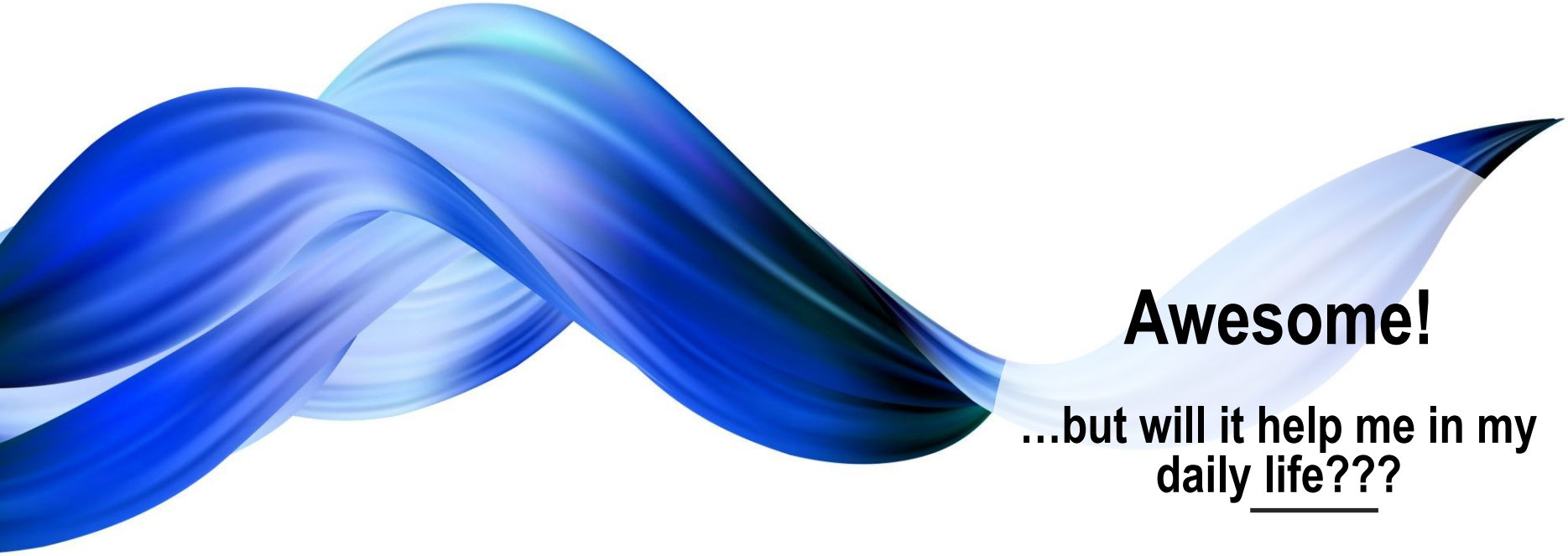
AI

Andrzej Wodecki, Midjourney
Gentle Woman

Learn from One Look (LOLNerf)

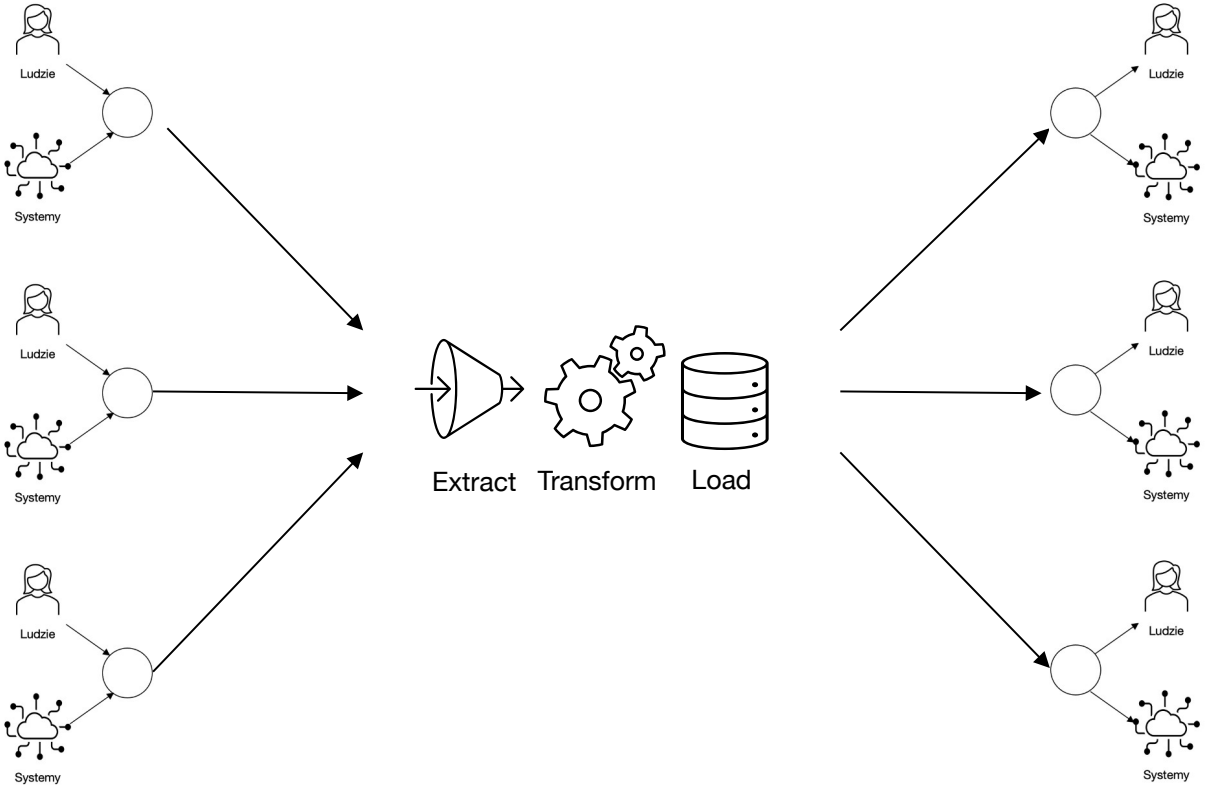

Prompt: An extreme close-up of an gray-
aired man with a beard in his 60s, he is deep
in thought pondering the history of the
niverse as he sits at a cafe in Paris, his eyes
ocus on people offscreen as they walk as he
ts mostly motionless, he is dressed in a wool
coat suit coat
vith a button-down shirt , he wears a brown
eret and glasses and has a very professorial
appearance, and the end he offers a subtle
losed-mouth smile as if he found the answer
to the mystery of life, the lighting is very
cinematic with the golden light and the
Parisian streets and city in the back

Stanford
University
0:04 / 2:37
HD



Awesome!

…but will it help me in my daily life???


Ludzie
Systemy
Ludzie
Systemy
Ludzie
Systemy
Extract
Transform
Load
Ludzie
Systemy
Ludzie
Systemy
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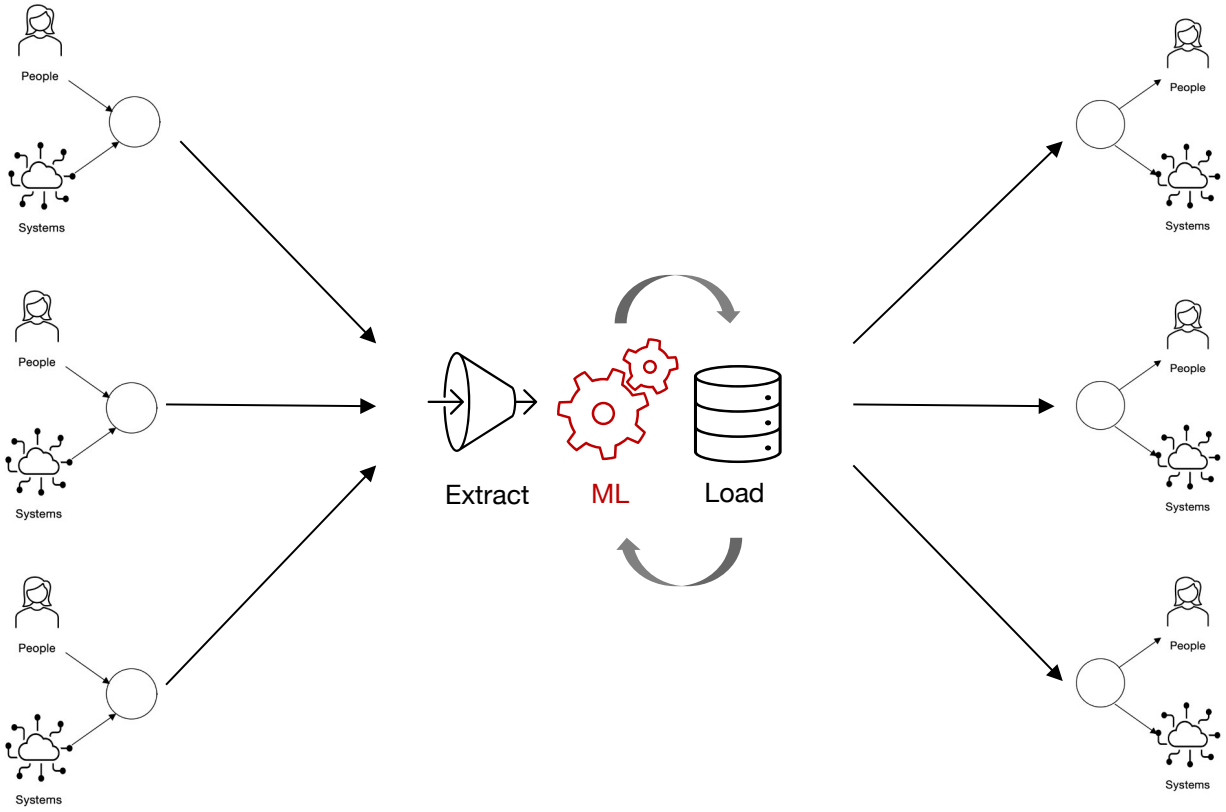
Systemy


People
Systems
People
Systems
People
Systems
Extract
ML
Load
People
Systems
People
Systems
People
Systems

What business questions can we find answers to in this dataset?

customerID	gender	age_first_order	user_agent_brand	user_agent_os	ip_address_country	ip_address_geopoint	campaign	pages_visited_avg	high_revenue
007494	F	43	Firefox	Windows	United States	POINT(-73.968 40.7588)	False	11.5	False
014834	M	48	Chrome	Windows	Peru	POINT(-77.0886 -12.0678)	False	9.5	False
024982	F	65	Firefox	Windows	France	POINT(1.9208 48.7501)	False	15.0	False
032754	F	52	Chrome	Windows	China	POINT(108.3167 22.8167)	True	9.0	False
040557	M	59	Chrome	Windows	China	POINT(109.8222 40.6522)	False	15.0	False
049591	F	72	Chrome	MacOS X	Ireland	POINT(-6.2439 53.3472)	False	11.0	False
058152	F	43	Edge	Windows	United States	POINT(-87.8945 43.0434)	False	11.5	False
067039	M	27	Firefox	Windows	Republic of Korea	POINT(126.9741 37.5112)	True	12.333333333333334	False
074702	M	43	Chrome	MacOS X	United States	POINT(-97.3478 32.9749)	True	9.666666666666666	False
082665	M	17	Chrome	MacOS X	United States	POINT(-110.3607 31.5273)	False	12.0	False
090266			Chrome	Windows	Czechia	POINT(14.4667 50.0833)	True	8.428571428571429	True
099664	F	27	Chrome	Windows	China	POINT(113.25 23.1167)	False	13.0	False
0pykn3	M	53	Chrome	MacOS X	South Africa	POINT(28.2294 -25.7069)	False	17.5	False
104603	M	48	Chrome	Windows	United States	POINT(-84.1789 39.7405)	False	11.5	False
111241	F		Chrome	Windows	United States	POINT(-93.8397 45.1845)	False	10.5	False
120957	M		Firefox	Windows	United States	POINT(-105.2037 40.0507)	False	18.0	False
131072	F	47	Chrome	Windows	France	POINT(2.4991 48.8702)	False	10.0	False
139844	F	37	Chrome	MacOS X	Australia	POINT(143.2104 -33.494)	False	15.0	False

What decisions can the answers to these questions help us make?

customerID	gender	age_first_order	user_agent_brand	user_agent_os	ip_address_country	ip_address_geopoint	campaign	pages_visited_avg	high_revenue
007494	F	43	Firefox	Windows	United States	POINT(-73.968 40.7588)	False	11.5	False
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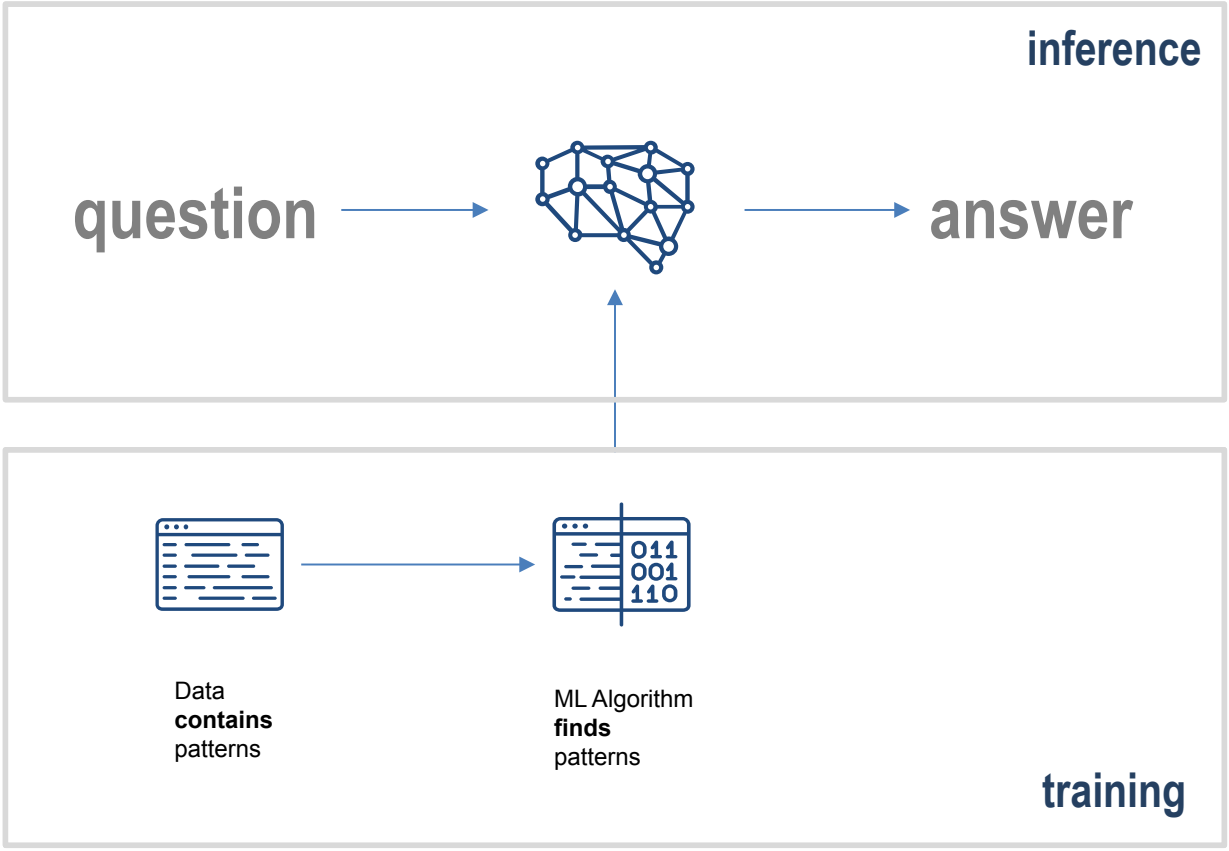
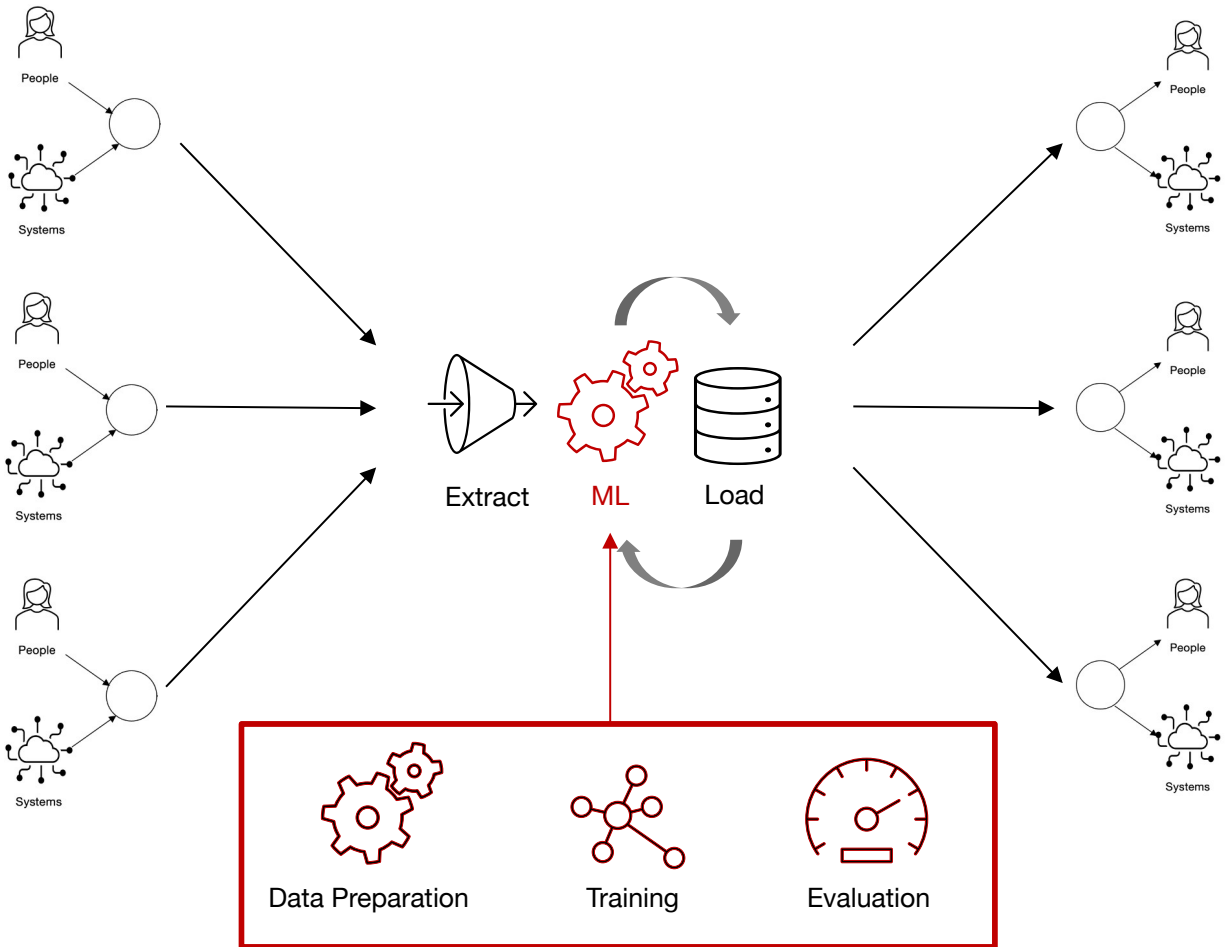
131072	F	47	Chrome	Windows	France	POINT(2.4991 48.8702)	False	10.0	False
139844	F	37	Chrome	MacOS X	Australia	POINT(143.2104 -33.494)	False	15.0	False

What knowledge and analytical skills should we have?

question → → answer


inference
question
answer
011
001
110
Data
contains
patterns
ML Algorithm
finds
patterns
training


People
Systems
People
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People
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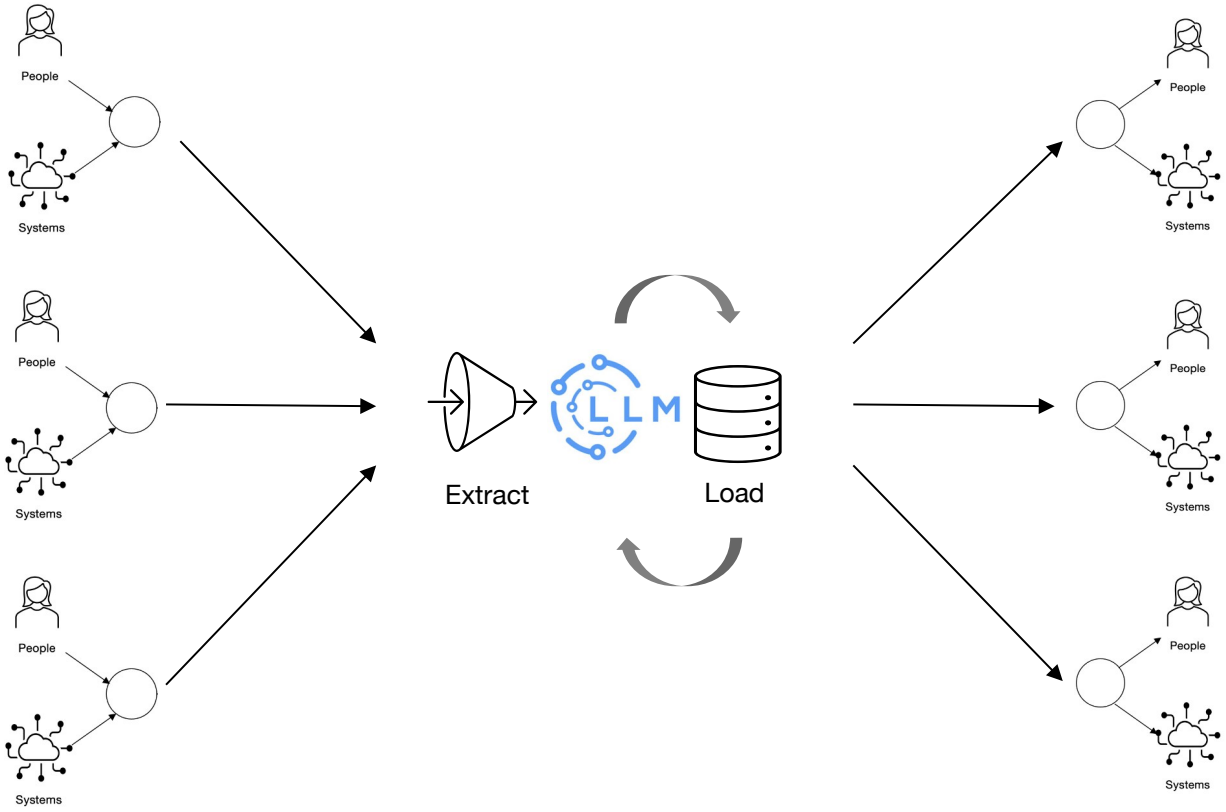
Extract
ML
Load
People
Systems
People
Systems
People
Systems
Data Preparation
Training
Evaluation

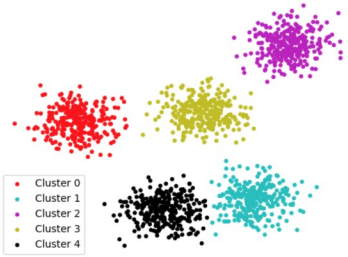

People
Systems
People
Systems
People
Systems
Extract
LLM
Load
People
Systems
People
Systems
People
Systems

classification

High/Low revenue?

Churn ?

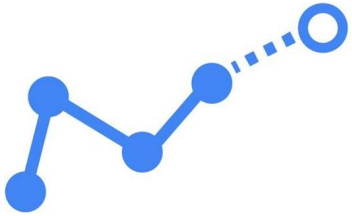
segmentation



RMF Segmentation

Marketing Campaign

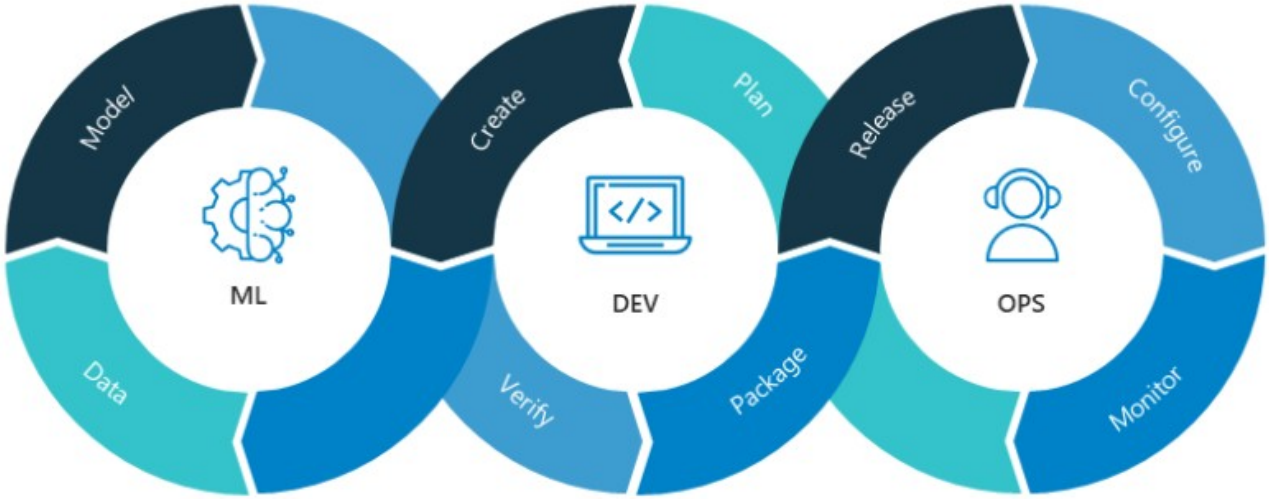
forecasting



Stock price

Demand

But in real life it’s a little bit more complex…


Model
Data
ML
Create
Verify
DEV
Plan
Package
Release
Monitor
OPS
Configure

ML / AI Infrastructure

DATA PREPARATION

Data Exploration & Processing
Paxata, TRIFACTA, alteryx, databricks, Qubole, HIVE

Data Version Control
Pachyderm, data iku

Feature Engineering and Storage
FEATURE LABS, Featuretools

Data Labeling
scale, Labelbox, figure eight, Amazon SageMaker, Google AI, Azure Machine Learning

Data Quality Checks
great_expectations

MODEL BUILDING

Hosted Notebooks Management
Amazon SageMaker, Azure Machine Learning, databricks, CLOUDERA, CO, DOMINO, Deepnote

Model Management, Version Tracking and Storage
databricks, Amazon SageMaker, Google AI, Azure Machine Learning, ALGORITHMIA, DOMINO, data iku

Experiment Tracking
Weights & Biases, comet, TensorBoard, mlflow

Model Optimization Hyper Parameter
SIGOPT, Weights & Biases, Amazon SageMaker, anyscale

Auto ML
H2O.ai, DataRobot, Amazon SageMaker, Google AI, Azure Machine Learning, data iku

Model Training
Amazon SageMaker, Azure Machine Learning, Google AI, Kubeflow, anyscale

Model Evaluation
TensorBoard

Model Explainability
fiddler, TensorBoard

PRODUCTION

Model Observability
arize, DataRobot, Other startups

Model Compliance & Audit
fiddler, Other startups

Model Deployment and Serving
Amazon SageMaker, Kubeflow, ALGORITHMIA, DataRobot, Google AI, Azure Machine Learning

Model Validation
arize, fiddler, Other startups

Platform Specific Model Builds
OctoML

Machine Learning tools & platforms landscape - v.1.0 January 2021

Presented by MLReef

DATA MANAGEMENT

Data Exploration & Management

COHESITY, rubrik, allegro.ai, ALLUXIO, Amundsen, druid, nuodi, MLReef, databricks, ALGORITHMIA, Spark, APARAVI, ATSCALE, amazon account, CAZENA, Google Cloud, CLOUDERA, kaggle, datagrok, data iku, ClearSky, Data Lake, DATERA, dremio, elastifile, erwin, Excelero, Fluree, GRAN, HYCU, imply, komprise, Parquet, pilosa, presto, tanx, VEARCH, vexata, YellowBrick

Data Labelling

appen, Dataturks, doccano, Dataiku, iMerit, prodigy, Labelbox, SUPERVISELY, Playment, scale, snorkel, HIVE

Data Streaming

ALLUXIO, nuodi, kafka, CONFLUENT, VALOHAI, Google Cloud, striim

Data Version Control

databricks, data iku, DVC, FLOYDHUB, MLReef, Pachyderm, Waterline Data, QRI, allegro.ai

Data Generation

scale, scrapinghub, DATPROF

Data Privacy

airlock, Celantur, mostly, PySyft, TUMULT

Data Quality Checks

arize, great_expectations, Naveego

MODELLING

Notebook / Code Management

colab, databricks, data iku, Deepnote, DOMINO, FLOYDHUB, Google Cloud, kaggle, MLReef, OpenML, polyaxon, Pachyderm, Weights & Biases, alteryx

Data Processing & Visualization

alteryx, ASCENDLIO, colab, DASK, data iku, databricks, dotData, Flyte, gluent, iguazio, imply, incorta, mlflow, MLReef, MODIN, Naveego, OpenML, Pachyderm, pilosa, presto, Prometheus, SAS, snorkel, SQLFlow, Starburst, vaex, VALOHAI, allegro.ai, Weights & Biases

Feature Engineering

dotData, FEAST, Featuretools, Pachyderm, tecton, SAS, data iku, iguazio

Model Training

alteryx, iguazio, Google Cloud, colab, databricks, data iku, DOMINO, dotscience, FLOYDHUB, Flyte, ludwig, kaggle, MLReef, mlflow, PerceptiLabs, snorkel, VALOHAI, SAS, Anyscale, Pachyderm, OpenML

Experiment Tracking

iguazio, allegro.ai, comet, data iku, DataRobot, datmo, DOMINO, FLOYDHUB, ludwig, mlflow, MLReef, OpenML, polyaxon, SPELL, VALOHAI, Google Cloud, Weights & Biases, Losswise

Model (Hyperparameter) Optimization

Amazon SageMaker, Angel, comet, DataRobot, polyaxon, SIGOPT, SPELL, tune, OPTUNA, alteryx, talos, allegro.ai, SAS

Auto ML

DataRobot, Determined AI, dotData, Google Cloud, kubeflow, TPOT, TransmogrifAI, iguazio, Amazon SageMaker

Model Management

alteryx, Amazon SageMaker, ALGORITHMIA, allegro.ai, databricks, data iku, Determined AI, dockerhub, DOMINO, dotData, FLOYDHUB, LUCIO, Google Cloud, OpenML, kubeflow, iguazio, MLReef, modzy, mlflow, PerceptiLabs, polyaxon, SAS, TUMULT, VALOHAI, Verta

Model Evaluation

arize, TensorBoard, MLPerf, Streamlit

Model Explainability

fiddler, InterpretML, LUCIO, PerceptiLabs, Shap, TensorBoard, Amazon SageMaker

Frameworks & major libraries

Chainer, mxnet, Keras, Spark MLlib, Caffe2, PyTorch, ONNX, TensorFlow, spaCy, XGBoost, CNTK, theano, matplotlib, fastai, julia, torch

CONTINUOUS DEPLOYMENT

Data Flow Management

ALLUXIO, Spark, ASCENDLIO, kafka, data iku, dotData, HYCU, PREFECT

Feature Transformation

FEAST, Featuretools, tecton, iguazio, data iku

Monitoring

ALGORITHMIA, arize, Google Cloud, data iku, DATADOG, DataRobot, DOMINO, iguazio, Losswise, snorkel, unravel, VALOHAI, Verta, kubeflow, datatron, Amazon SageMaker, alteryx

Model Compliance & Audit

ALGORITHMIA, SAS, kubeflow

Model Deployment & Serving

AIBLE, ALGORITHMIA, allegro.ai, cortex, data iku, datatron, datmo, DOMINO, dotData, FLOYDHUB, FRITZ AI, iguazio, Kubeflow, mlflow, modzy, OpenML, PaperSpace, PredictionIO, SAS, SELDON, SPELL, Streamlit, Amazon SageMaker, VALOHAI, Verta, kubeflow, Google Cloud, Azure, Amazon SageMaker, alteryx

Model Validation

arize, datatron, fiddler, LUCIO, MLPerf, SAS, Streamlit

Model Compatibility

MMdnn, ONNX, OpenDVL

COMPUTING MANAGEMENT

Computing & Data Infrastructure

Google Cloud, Azure, Amazon SageMaker, CLOUDERA, ORACLE, Amazon EC2, linode, DigitalOcean, Linux Foundation, Docker

Environment Management

CONDA, databricks, datmo, MAHOUT, MLReef, allegro.ai

Resource Allocation

ALGORITHMIA, Google Cloud, MLReef, Azure, databricks, data iku, Determined AI, FLOYDHUB, Amazon SageMaker, polyaxon, SPELL, VALOHAI, allegro.ai

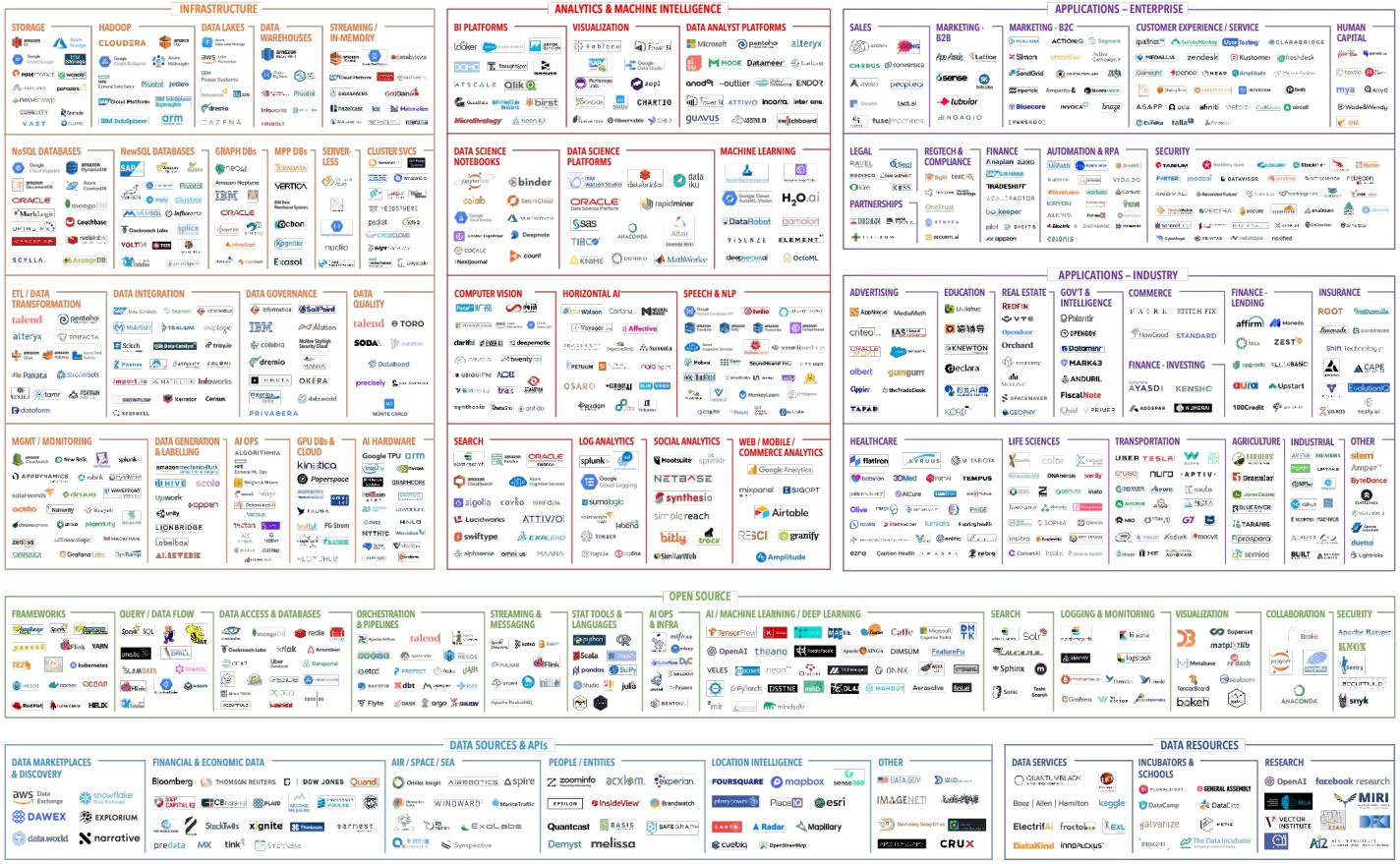
SCALING

orgo, Azure, DATADOG, datatron, datmo, Google Cloud, Amazon EC2, SELDON, tvm, kubeflow

SECURITY & PRIVACY

Azure, DATADOG, modzy, PySyft, TUMULT, ALGORITHMIA

DATA & AI LANDSCAPE 2020

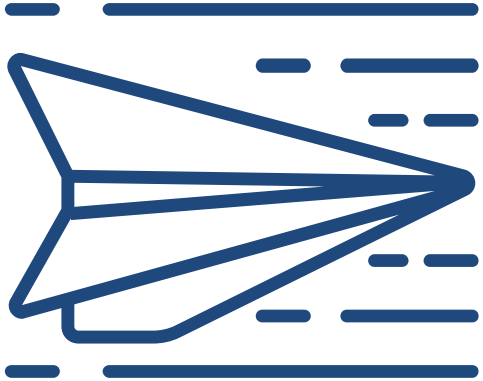


© Matt Turck (@mattturck) & FirstMark (@firstmarkcap)

mattturck.com/data2020

FIRSTMARK
EARLY STAGE VENTURE CAPITAL

This landscape is rapidly changing



What should I know then?

It's much more important to know now what to do (in a given situation), rather than How to do it

Stick to the most popular, growing libraries
{Look at the GitHub commits stats}

Don't avoid commercial solutions... They may indeed be cheaper

... but build Your foundations with open source
It's harder, but gives you a sense of what's going on under the hood

MODERN DATA SCIENTIST

Data Scientist, the sexiest job of 21th century requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- Machine learning
- Statistical modeling
- Experiment design
- Bayesian inference
- Supervised learning: decision trees, random forests, logistic regression
- Unsupervised learning: clustering, dimensionality reduction
- Optimization: gradient descent and variants

PROGRAMMING & DATABASE

- Computer science fundamentals
- Scripting language e.g. Python
- Statistical computing package e.g. R
- Databases SQL and NoSQL
- Relational algebra
- Parallel databases and parallel query processing
- MapReduce concepts
- Hadoop and Hive/Pig
- Custom reducers
- Experience with xaaS like AWS

DOMAIN KNOWLEDGE & SOFT SKILLS

- Passionate about the business
- Curious about data
- Influence without authority
- Hacker mindset
- Problem solver
- Strategic, proactive, creative, innovative and collaborative

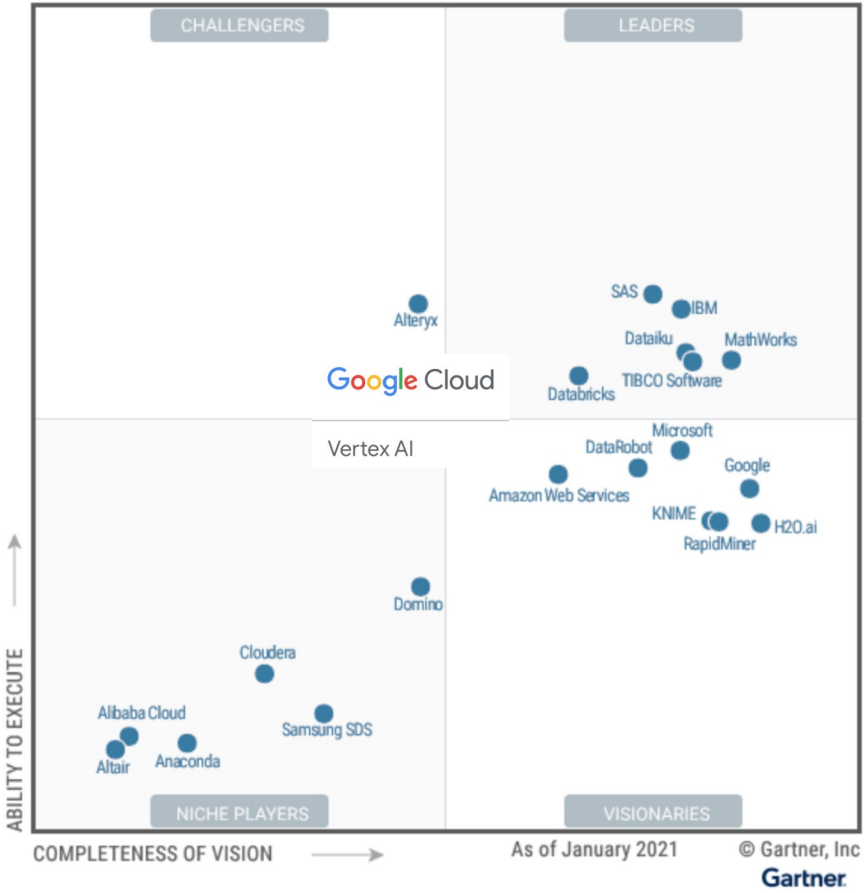
COMMUNICATION & VISUALIZATION

- Able to engage with senior management
- Story telling skills
- Translate data-driven insights into decisions and actions
- Visual art design
- R packages like ggplot or lattice
- Knowledge of any of visualization tools e.g. Flare, D3.js, Tableau

MarketingDistillery.com is a group of practitioners in the area of e-commerce marketing. Our fields of expertise include: marketing strategy and optimization: customer tracking and on-site analytics: predictive analytics and econometrics: data warehousing and big data systems: marketing channel insights in Paid Search, SEO, Social, CRM and brand.

Marketing DISTILLERY

Figure 1: Magic Quadrant for Data Science and Machine Learning Platforms



Source: Gartner (March 2021)

https://www.dataiku.com

The Gartner 2019 Magic Quadrant for Data Science and Machine-Learning Platforms

Get the Report

data iku

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Your Path to Enterprise AI

Bring data analysts, engineers, and scientists together. Enable self-service analytics and operationalize machine learning. Get results today and build for tomorrow.

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Dataiku Named a "Challenger" in Gartner 2019 Magic Quadrant for Data Science and Machine-Learning Platforms Gartner

PFIZER
Developing a Platform for Collaborative Analytics

BCG
People-Driven Approach to Analytics at Scale

EGG CONFERENCE
Inspiration from Leaders in Analytics, ML, & AI

GE | Santander | KUKA | FOX | paloalto | L'ORÉAL | Unilever

Final Assignment

Lab: Project

Exam = electronic test
@PJATK

Final project

Title

Author(s). Individual, or a group project (up to 4 persons in a group)

Problem descrition

A brief description of the problem and its context (industry, environment, etc.)
Who can benefit from this model and how?
Why does the problem seem interesting to you?

Data

Sources of data, assessment of their reliability
Brief descriptive analysis of the data
Justification: how can this data help solve the problem?

Solution

Brief description of the chosen approach with justification
Project stages
Model evaluation aproach (measures, and strategy)

Model evaluation and discussion

Model evaluation
Discussion

Summary

What was the goal of the project?
Who and how can benefit?
What are the models limitation?

Course materials

https://bit.ly/PUM_ENG2024

wodecki@pjwstk.edu.pl